
Improving The Police Force Control Rooms Efficiency Using Simulation Analysis

Executive summary

This report describes a consultancy project carried out by Lancaster University to The Police Force (TPF) in order to improve control room operations. TPF receives approximately 2 million calls every year and makes about 800,000 outgoing calls. Currently TPF is experiencing these problems:

- Public waiting times do not meet performance targets in emergency and non-emergency situations
- Concerns about police reaction time and arrivals to incidents
- Excessive overtime bills

In order to find possible ways to improve the current situation, simulation was used as previous practices showed good use of it. First, current operations were analysed and observed with help from TPF staff. It was clear that contact and response centres (CaRCs) were made of two parts:

1. The front end – where call takers would take calls from the public, offer advice and grade each situation according to its urgency
2. The back end – where radio operators would get information from call takers and would assign police resources to different cases

Only front end operations were simulated. Surprisingly, two issues arose while gathering data for the simulation model. First, staffing was not managed very strictly and shift patterns did not reflect the true situation and absenteeism was a problem. Moreover, while investigating back-end operations, it was clear that radio operators had little control over resource deployment, as most calls were processed in any convenient order by police officers. Moreover, it was clear that current operations of CaRCs consisted of multiple queues for each server, which in practice is not efficient.

Two different simulation models were built: as-is simulation model (with 4 different queues for each CaRC) and proposed model with single queue. Results showed that the as-is model reflects the current situation accurately, so these two models were compared. The proposed model outperformed the current model on all aspects. Moreover, the proposed model was meeting current TPF answering call targets. Later the proposed model was tested on stress conditions (increased call durations and stricter answer time targets). It still performed very well and managed to meet targets.

It was clear from simulation that the current operational system will not meet targets, which might be caused by poor management of staff and an inefficient system. Also, the current logic of trying to integrate call handling and resource dispatch within the CaRC system did not make sense. Moreover, an improved system with only one queue should be used as from the simulation model it was clear that it performs better. Additionally, stress analysis of the proposed system showed that current staff levels would be sufficient enough for increased call durations and tighter answering time targets. Thus the following recommendations were made:

- Locate radio operators in subdivisions
- Replace 4 CaRCs with 2 Emergency Call Centres (radio operations removed)
- Replace current 4 queues system with single queue system

If applied, this could lead to improved control of radio operators, more consistent work for front-end operations (call takers), improved efficiency and achieved performance targets.

Project Background

The Police Force (TPF) is currently serving 500 square miles and receives 2 million incoming calls every year, correspondingly making 800,000 outgoing calls. All calls are grouped into emergency calls (999 calls which are redirected by BT operators to TPF) and non-emergency calls (calls coming from Single Non-Emergency Number (SNEN)). These calls are assessed on performance targets (currently set by TPF) indicated below:

Call type	Performance indicator	Target (%)
Emergency (999 calls)	Percentage of calls answered in 15s	90
Non-emergency (SNEN* calls)	Percentage of calls answered in 30s	90

Table 1

TPF is currently struggling to meet set targets. Moreover, Public Service Agreements for the Home Office have included stricter targets for reduced crime and increased public confidence which are driven by concerns about police reaction time and arrivals to incidents. TPF is also incurring large overtime bills that are thought to be excessive. This lead to the following objectives:

- Analyse current contact and response system
- Find possible ways to improve it
- Check if improved system meets stricter targets and reduces overtime bills (which would lead to better performance of TPF)

Approach

Simulation modelling was used to help with this problem. First of all, Operational Research literature suggests that it is one of the best approaches in dealing with similar problems. Secondly, TPF has Micro Saint Sharp software which can be used for making simulation models of entities passing through task network. However, it has to be noted that successful work depends on close cooperation of the analyst team and the client. Thus, it has to be pointed that simulation modelling has its limitations:

- Good understanding of the conceptual model is needed, which requires close co-operation with TPF
- Data has to be collected for inputs of the simulation model
- Simulation models is just simplification of real world situations
- Simulation models have to be tested to see if they represent real world situations accurately

On the other hand, simulation models have many advantages, which influenced the decision of using simulation models greatly:

- Once built, simulation models can be modified to see how the system would change in the real world
- It is much cheaper to implement changes on the simulation model than in real world tests
- Changes can be observed much faster than in the real world

Thus, it was decided to first build the “as-is” model and then proceed on making changes in order to find ways to improve the current system.

Analysis

Constructing the current models requires good understanding of the system. In order to achieve this, the conceptual model was built with great help from TPF staff. The conceptual model building was carried out as an iterative process by interviewing staff and asking them whether our conceptual model represents a real-life system accurately. After mutual understanding and agreement on the conceptual model was reached, we started to build a simulation model. It is important to note that building a conceptual model not only helps with further development of the simulation model, but also helps TPF staff gain better understanding of the way they operate.

From the conceptual model it was clear that the current Contact and Response Centre (CaRC) comprises of front-end and back-end parts:

- The front-end: call takers graded each call to produce an Incident Report (IR) by typing appropriate information into a pro-forma while speaking to the caller. The support software entered IRs into a database and then ensured that a summary of the IR appeared on the screen of a radio operator (back-end operations).
- The back-end: the radio operators viewed the IRs on their screens and prioritised them by their call grade, dispatching appropriate resources. On some occasions, the radio operators, assisted by their supervisors, had to further prioritise calls for two reasons: some calls might have the same call grade awaiting resolution and incidents being reported through multiple calls.

System analysis and data gathering lead to worrying findings. First, TPF staff was ill-informed about how Automatic Call Distributor (ACD) worked. Secondly, after running the initial simulation model it produced completely different outputs compared to real-life data. After further investigation, it was found that staffing schedules do not represent true staffing levels. This was mainly caused by the fact that the call-takers often substitute radio-operators, in other words, there is a clear management problem of staff, which also had an impact on absenteeism and overtime work. Thirdly, radio operators had little control of resource deployment, as the most often subdivisions of TPF would assign appropriate police resources. Although this might be preferable, as subdivisions have better local knowledge, this means that the current system is not very efficient. Lastly, it was clear that having multiple queues for CaRCs is in practice inefficient. It was decided to focus on front-end operations as it was clear that simulating back-end operations was infeasible.

After all data has been collected the current system simulation model with 4 queues for each CaRC was constructed. It was checked to see if it represents a real-world situation accurately. Also it was tested to see how it performs on target levels pointed out before.

As from initial analysis it was clear that having multiple queues is inefficient so the improved model was built. It consisted of a single queue, 4 CaRCs were replaced by 2 Emergency Call Centres (radio operations removed (ECC)). Moreover, radio operations were removed, because it was clear that call-takers and radio-operators communicate only via database, which means that they could be located in different places. This model was also tested to see how it performed against performance targets.

Results

From running the simulation on the “as-is” model it was clear that performance targets could be met if staff was managed properly and call-takers would not substitute for radio-operators; although stricter targets would not. However, it was clear that the current system is inefficient in different aspects. Moreover, trying this model on stressed conditions showed that staffing levels would have to be increased in order to meet the performance targets.

The improved model proved to be much more efficient, meeting all the performance targets. Moreover, it performed well on stressed conditions and stricter performance targets. It was clear that the current staffing levels would be enough with the improved model even on stressed conditions.

All results calculated as percentage of calls answered within target times are summarised in the following table:

	Emergency calls		Emergency calls	
	Current system	Improved system	Current system	Improved system
Average	94%	99%	86%	99.80%
Best Weekly	98%	100%	93%	100%
Worst Weekly	86%	97%	72%	99%

Table 2

From Table 2 it is clear that the improved system performs above target performance levels even on the worst weekly percentages. Moreover, it is clear that separating radio operators and basing them in subdivisions would lead to improved staff management. Also having a single queue system would mean that work for call-takers would be more consistent, with less fluctuation of number of calls during peak/off-peak times.

Reorganisation of the 4 CaRCs into 2 ECCs with dispatch devolved to sub-divisions will be expensive and will need to provide robust performance even if call volumes or durations increase. Thus, the long-term implications were also considered. The improved model was tested on a 15 year model of constantly increased call durations. The experiment showed that the improved system should be able to meet current performance targets without increase of staffing levels. Analysis on increasing call volumes showed that an improved system would work well for the first 10 years and later staffing levels should be increased.

However, it should be pointed out that back-end operations were not analysed in great detail and changes of how police resources are allocated could be considered in further analysis. Focusing only on one side of a system may switch the pressure elsewhere and it is important to ensure that meeting call handling targets would lead to meeting of response targets.

Conclusion

From the above analysis it is clear that the current 4 CaRCs system is inefficient and TPF suffers from poor management of staff. This causes higher overtime bills and poor performance of call centres. The main problem causing inefficiency is 4 queues, which could be replaced with a single queue system, radio-operators could be relocated to subdivisions (as call-takers and radio-operators communicate via database) and 4 CaRCs could be replaced by 1 ECC. This would lead to better performance, better staff management of radio-operators and robustness for long-term operations.

Recommendations

Having all that in mind it is recommended to:

- Locate radio operators in subdivisions
- Replace 4 CaRCs with 2 ECCs
- Replace current 4 queues system with single queue system

If these changes were applied, TPF could meet even stricter performance targets which were proposed by Public Service Agreements for the Home Office. Moreover, the improved system would be time robust which would outweigh costs of replacing the current system and re-modifying call-centres' operations.